

ActInf Livestream #021.04 ~ John Boik

Source: [YouTube](#) **Playlist:** John Boik ~ Livestreams on "Science-Driven Societal Transformation" **Duration:** 1:54:38 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome welcome to actin flab we're in live stream number 21.04 on april 27th 2021 and today we are in the fourth of a series of conversations with john boyk so john it's been really awesome to get to know about you and your ideas and work with you on these jamboards so as i sort of set things up for a second and share the jamboard link with the live chat go ahead and just re re-catch us up and then i'll meet you on the first slide in just a minute okay sounds good thanks uh so my name is john boyk i'm a courtesy faculty at oregon state university um and the series of discussions that we've been having is on the is on a paper a series of papers that i just published in the journal um sustainability it's a three-part series the topic is well the title is science driven societal transformation parts one two and three so uh we've this is our fourth get together here and we're just finishing up our discussion of the last in that series part three which which is uh the subtitle design and um we we covered quite a bit in that paper in that paper part three but now today we're going to focus on just one aspect of that which is uh the leader framework um local economic direct democracy association framework this is a what i'm i'm terming it a prototype of design for new societal systems and as a reminder to listeners we're talking about the the de novo design of new societal using specifically club model so the societal systems for a club the club is a is a small community somewhere something larger than say a thousand volunteer participants that's how the in the art the series itself proposes an r d program and that's uh the strategy for change in the r d program is new systems are tested at the club level the community level in cooperation with the global scientific community who's working on the technical aspects then field trials occur at the club level and if the if the new systems deliver as promised and they promise substantial improvement and uh social and environmental quality economic security and various things like that then the idea is that the club model would grow and spread and describe to new locations and uh new testing and new trials and eventually would spread near on a near global basis so are we on slide one yet we're ready on slide one okay so that's our that's just our uh our list of discussions we've had you can check out the links there for uh the previous youtube

videos and slide two if you would go there is uh just where just where to go for more information so the the my website for this project is called principal-societiesproject.org and it uh lists the uh gives links to the series get some backgrounds information um actually has a it on there is a live or interactive model of the leader framework so so part of what will be a big part of what we'll be talking about today is actually on the website if you want to run the model yourself so they go there for more information and then third slide so today's topic again is the leader local economic director democracy association these are ideas from my 2014 book called economic direct democracy that book is available for free on the website or if you want to spend money you can go to amazon and buy it there [Music] and along with that book i published a simulation model of leader framework or at least parts of the little leader framework and that is in the international journal of community currency research so links are there for that and again all of these links are available on the principal societies project website okay so now why would i want to discuss this is slide 4 why would i want to discuss the leader framework in this three-part series a few reasons one is that the leader the design of the leader framework maybe should say that the leader framework is intended as an integrated societal systems architecture or what i've been calling a cognitive architecture the the idea behind this whole series behind this work is that a society large or small is a cognitive organism and its societal systems which for example economic systems legal systems governance systems etc act as a cognitive architecture so so when discussing the idea of societal transformation my focus here is on uh creating new societal systems de novo societal systems design uh so that they function uh as cognitive systems they facilitate societal cognition and and this is a fitness-based evidence-based r d project so the idea would be that you could assess how well some did one how well one design is doing over another design and eventually communities could choose designs that function very well and are fit for purpose all right so the lida framework embodies the cognitive viewpoint it also was designed uh to be implemented via the club model which i talked about in part two of the series that's the strategy for change um number three is here the reason is that it involves integrated systems so uh there's i in the in the series i talk about six overarching systems legal governance etc and the the leader framework uh integrates most of them and finally there's a published simulation model available at least for the economic some economic aspects of the and at the time and i think that still might be true this is the first semi-realistic simulation of a de novo system a de novo economic system that i'm aware aware of number five all right so i'll get into the we'll talk a lot about the simulation today and a little bit about the you know the the uh structure of

the but let me just offer a few warm-up thoughts uh um the the lita economy is not like what we face a normal economy that's it's it's fundamentally different and in particular it is a democratic uh it is a cognitive architecture as i've said and it is deeply democratic and transparent so in the in the lead economy money there is money both the national currency in america the dollar and a local digital currency that i call the token so those two currencies are used as a voting tool in a highly transparent fair and deeply democratic process and as we talked about last time money is already used as a voting tool in our world it's just that it is a non-transparent non-democratic voting tool for the most part there's some people have billions of votes in the economic system and then other people have just you know a small number of votes so uh the leader framework makes that voting function explicit and fair and democratic the uh the emphasis of the simulation we'll be talking about in a moment is on the economic system but as i mentioned the leader framework encompasses the you know all or almost all six um overarching systems i i this this simulation is intended as a first step and uh it is it's been it could be made more sophisticated and uh this so this first one simply looks at the economic aspects aspects because that was the simplest to do for me at the time um so money is used as a voting tool and it's and voting occurs in two arenas it occurs in local markets so i buy some product from from a local store for local business and it also occurs in in the financial system what i call the crowd-based financial system and we'll talk a lot about that but essentially individuals contribute a certain amount of their income to the crowd-based financial system and then once their money is in the crowd-based financial system the cbfs then they have uh they get to choose how that money is used so in a sense they get to vote for they give and offer their votes for how the community is what the community wants what the community wants to fund how what kind of jobs the community wants to fund and so on um so if you had a if you had an economic system in which money was used as a voting tool well uh it would be a fairly undemocratic unless incomes and wealth were somehow equal so over time the effect of the leader is to equalize incoming wealth and i'll show the show the results of that in a minute the the latest simulation runs 28 years and as you'll see at the end of that 20-year period everyone who has joined the club uh is enjoying an equal level of an equal and high level of income and wealth and uh lastly the leader framework is designed to focus on motivations that is the any new system developed in this r d program the goal is to have that system be fit for purpose and as we've talked about the purpose of a society the purpose of societal cognition and the purpose of i should say the implicit person purpose of societal systems is to facilitate societal cognition and to and thus to ensure that the organism the

society maintain achieves and maintains a high degree of vitality so the any new systems developed should be fit for purpose and and if they're fit for purpose almost by definition they will facilitate cooperation because cooperation and communication is a part of cognition so the lead of framework and what rewards barriers to cooperation and with the idea that that it makes cooperation so easy that cooperation becomes the default behavior and it also focuses on motivations it does not reward people for uh non-cooperative behavior for example and that is in contrast to our existing systems which which i call native systems in the series but i in our in native systems i am for uh paying my workers a very long for the society would reward me greatly for that and that doesn't occur in the partly because income and wealth are equalized over time thus the the part the motivation to participate in the economic system of elita or we'll call it we can also call it a club model is not personal gain but community well-being well-being of the larger organisms are the members of a club are you thinking uh spatially co-located are they remotely located yes locally in space and it is it was perhaps possible to have a club that is distributed over over space but um i didn't start that way i mean that maybe could be something that's added later but i'm focusing now it's the easiest thing for me has been the easiest thing to me for me to focus on the local market local area but there's reasons why the local area is important because many decisions actually are local like who will pick up my garbage and where will that garbage go is a local decision you think the many of the decisions many of the of the our quality of life is largely a local issue for example our my economic security is largely a local issue and obviously being close in space allows a greater opportunity for community building and gathering and seeing face to face and communicating communicating face to face and getting to know each other and building a deep sense of friendship and community that way but yes in the future communities might be extended in some way and as a matter of fact uh the the idea with the leader framework or the idea with the club model is that that the initial focus would be on developing the tools and systems for clubs for the project and then that would land systems for club networks so that would be by definition non-local you know non-local engagement because clubs might be in different areas of the of the world okay uh so is that right yep um so there's in the bottom left of that slide over in the bottom left that slide i'm listing the six overarching systems that i talk about in the in the series and you can you will see that all but leader so i would i would imagine you know that if this were developed further that health would also be included but i just haven't included it yet uh so very briefly the sec this uh seven integrated components the first three i'm calling uh i call them collectively the token exchange system again the token is the local

currency and it and it travels in parallel with the national currency with the dollar locally uh there's a token monetary system that creates and destroys tokens there's the crowd-based financial system which i mentioned already that uh that to be clear that cbfs handles both the national currency and the token it issues to local businesses and non-profits uh it issues donations interest-free loans and subsidies and these are all transparent through democratic decision making and it also has a nurture arm that people contribute to and members contribute to and that nurture arm provides income for not what i call not in workforce members so what when i say that the leader achieves income equality over time i mean for every single a member so you can think of it as every member family or however you want to think of it but uh regardless of whether that person is in the workforce not in the workforce ill not ill employed not employed in that sense it's it's there's it's a little bit like a universal income only it's a very high income same income that everyone is enjoying i'll show you that in just a minute and then number three in the in the tes token exchange system is the market so these would be local local markets that are better members and uh part of the some of the new businesses that are developed by the leader would be called principal businesses principal businesses and that's a very special business model unique to the leader kind of like a non-profit kind of like a hybrid between a non-profit and a and a regular for-profit business um number four there's a property rights system that goes along with this for example a club a leader would have an ip pool for intellectual property and the and the club could also manage other properties could manage real estate properties for example or any kind of property so it would be democratic decision making you know for property management also there'd be some restrictions on the sale of principal businesses kind of similar to non-profits and that would be high transparency throughout so the community would know if the community is using money as a voting tool the community would need to know where their what's happening with their votes what's what was the you know what are the choices that the community is making with with money and so the flow of money would be highly transparent compared to today um number five is an information and incentive system this would include for example education programs reputation system the fitness indexes that we've talked about which allow the community to understand itself how it's doing what's going to happen next uh how how what is the quality of the design that we're using could we improve our design uh and along with fitness indexes the last one there's computer and other kinds of forecasting and assessment methods that would help a community to understand what are the trends and how is the system working and how are we doing um number six is the collaborative governance

system and that has three branches administrative legislative and judicial so that then covers the legal system and finally number seven is purpose and we've already talked in each of these discussions we've talked about the the importance of purpose and how purpose purpose follows world view and um and and design and metrics follow from purpose okay so next okay all right so now we'll we'll spend most of the rest of this uh discussion on the simulation model because i think that'll that'll be most that'll make the description of the leader foreign and again uh the the this is this is a first simulation um it's it's intended to introduce the the um and also just to describe some general concepts of token dollar i'll use dollar as the national currency if you don't mind token dollar flow and um and it's the third aim of this uh on the left paragraph there is to demonstrate at least uh you know conceptually that income equality can be achieved um that there's some parameterization of this model that you some you can choose parameters such that uh this goal of income equality can be achieved at the end of the simulation the the simulation lasts for about 28 years i believe um so this is not a this is not uh you know this is the the the concept behind the leda is that it bootstraps its success it starts small it grows from there it injects like in the very first year maybe just a few members will join and the leader injects a small amount of tokens as the as the local currency into the economy and makes some very small changes and then the next year there's another few people join it makes some bigger changes the next year makes another few people join and make some bigger changes still and the the growth of the leader occurs exponentially fast now i would like i would like to say that the functions of the leader how it how it works how it achieves these aims how it achieves income equality for example this would all be worked out far in advance before any field trial ever occurred so what would actually happen is there would be say you know who knows hundreds or more of simulations or papers or at least let's say a large number of simulation and papers on a variety of topics would be finished before any community was ever asked to participate in this so once that body of knowledge is developed then perhaps the the the program would say okay what is there are there any communities in the world that would be interested in participating in a field trial and if you did participate here's the list of benefits that you might expect over if this say a two-year field trial uh income levels will increase this is what's going to happen this is this is how participation will increase this this is you know this the the trajectory would be laid out through a variety of simulations so everyone should know what they're getting involved in what they expect from this and and there should be no surprises from the field trials now that doesn't mean that the community has no choice in their behaviors and what they do with the system they the

community drives the system but the system is designed such to achieve its purpose right fit for purpose um all right so um um on the left there i just want to emphasize this that this simulation conveys broad low level low resolution design and intentions for an actual token exchange system so i'm not predicting that a token exchange system would work like this this is rather a illustration of the concept of the token exchange system in the leader framework and obviously this simulation can and hopefully will be followed by a large number of other simulations that are progressively more sophisticated this is a in many respects a fairly simple simulation and there is tremendous room to expand it and improve it um but it's still useful and in particular that's because i made the simulations stock flow consistent which means that no no monies come out of thin air it's all you know if money leaves one agent then another agent receives it so all the flows of money are accounted for and just that alone allows one to say that okay logically you know logic mathematically this all adds up um and i would lastly i would say that this is a what i call a semi-realistic simulation that is uh income level starting income levels resemble real income levels tax levels tax rates resemble real tax rates thus the initial conditions are some silly realistic but but again the simulation is very simple so i start with a semi-realistic initial setting and then certain things don't change over the years it's just sort of the only thing that actually changes is just what is pertains to the lead itself the members themselves so if you don't join and meet in this in the simulation if you don't join the leader over the period of years then your income stays just what it was and um that nothing really changed just for you you keep the same job and keep the same family and things just move ahead all right yeah please okay you could address it uh now or later but what is the relationship between your simulation and game theory and what is the relationship between your simulation and active inference oh okay um okay so um in a this is an agent-based so in that means uh you know between the lines that means that there are individual agents that make decisions of what they want to do and that determines how the the history of the simulation goes um in a in a highly sophisticated agent-based models the agents themselves might be very complex and they might make decisions about complex situations and since this is a just a simple introductory model the agents are relatively simple um like for example a family and they make fairly simple decisions and the main decision they make in this uh simulation is whether or not to become members in any given year does a family want to become a member secondly they'll make a decision on how they what they choose a wage option which is a construct that i'll talk about in a so there's not very many decisions that uh that individuals will make here so it is like a game theory but it's a very simple game and later

simulations could be made much more sophisticated and in relation to active inference i obviously i might say that i created these the book and the simulation model long before i was aware of what active inference was so i what the book neither the book nor the simulation model mentions the cognitive aspects and that came a little bit later in my in my development and as the project developed in my mind but but one could easily imagine how active influence inference might fit into this for example you know as i said a a is considered to be a organism it's understood as an organism a so just like every organism it has to fulfill its intrinsic purpose which is to maintain and achieve vitality and it and it in the active inference scheme it um it assess it it receives information from its world through its senses it updates its models it chooses actions that changes the world and then it receives new sensations based on the changed world and perhaps updates its further or chooses a different set of actions or continues with the same action but but that's how active inference works for an organism but that's how active inference would work for a community too the community senses information it decides you know it learns what it can learn from the current uh sensations it makes decisions about what it'll do next in a variety of ways what will it do with its solid waste where how will it generate energy what will it fund next uh what what businesses will it fund what nonprofits will it fund what is it what is its targets does it adjust its targets so these are all decisions that uh that organism of a community would make as it tries to to achieve and maintain vitality and fortunately uh since this is a fitness-based program the community would know at every moment how it's doing what it's what its indexes are it's you know it's indexes for vitality for example or it's or it's indexes for uncertainty about the future so as it tries as the community tries to reduce uncertainty about its essential variables then you know it would have an immediate feedback as far as like scores about how it's doing and those scores and that information could help it then to update its models or uh and choose the same or different actions and so that's the that's how active inference might fit in in the community as a whole but you could imagine that any you could focus in on any individual part of this stay just a principal business and the principal businesses the principal business could also be thinking about its cognition in terms of active inference it too gathers information and then makes uh makes adjustments and learns and chooses actions and another example is the cloud-based financial system itself this is a decision-making organization structure and that whole structure could be modeled using ideas from active inference cool it's interesting how you developed and applied basically the evolution of computational modeling to it and that helped a few things maybe come out into relief like that's that multi-scale aspect and then also the fact that

you're pursuing not the maximum gdp you know not the the total divided by the number of people which you're maybe going to get a higher average with one billionaire and everyone else has zero but that's not the pursuit it's actually a uncertainty minimizing trajectory policy sequence that's starting to sound a lot like active inference so it's kind of cool how it dovetailed together right right this is these are these are you know these are probabilistic this you could say the community is developing a probabilistic model of what will happen next what will happen to me and uh and that fits in very well with the active inference scheme and individuals in the community are developing a generative model of of how i see the world and what will happen next and as a matter of fact since the that since it's so important since anticipation is so important in cognition the individuals in the community individual organizations such as businesses and the community as a whole would be developing anticipatory models whether those are formal models uh you know in computers or whether they're informal models in their in their minds about how how i see the world and how i see what will happen next if a b and c happens so all of that information that full anticipatory capacity of individuals and organizations and the community as a whole can be taken taken into account in the leader can be used in the leader for the for its own good all right number uh slide eight yep we're there okay all right so um this uh this image on the right really captures much of the much of the simulation and much of the idea of the leader as far as the token exchange system goes um and before i describe it i just want to emphasize once again that the token exchange system is just one part of the leader and there's other parts that are not up not in this simulation at all for example the governance governance system is not in the simulation at all so this is just flows of currency and uh figure one on the right describes how that works to to simplify this a little bit let's just look at the arrows in blue and the arrows in blue are both tokens and dollars that flow together and we could start at organizations these are the agents of the model organizing the five agents organizations persons cbfs rest of counties which is the rest of the world and government which is local government um just looking at the blue lines we see there's kind of a circle being made between organizations persons cb cbfs and then again to organizations and that's essentially how how how things flow organizations persons are employed at organizations and organizations pay wages to persons persons spend money back into organizations when they buy goods and services persons support the cbfs with contributions and the cbf fs supports persons in particular not in workforce persons with what i call nurture support the cbs also provides loans subsidies and donations to organizations and the organizations pay back the loan principal i might add that the the concept of loans here

is uh is the token itself is designed to be a non-inflationary uh uh um currency and loans are offered interest-free so what you don't see here in this circle i just described what you don't see is investment for profit that does not occur in the system um uh especially with relation to principal businesses it doesn't occur and obviously doesn't occur with non-profits because non-profits are by definition not for profit so so if i'm a person i receive wages from my employer or maybe i'm self-employed i sell things to the other and i have to contribute a certain fraction of my income to the cbfs and once it's in the cbfs i can organizations organizations will apply to the cpfs for loans or subsidies or donations and i along with my fellow colleagues in the i have a pool i have an account there and i can decide you know i like i like that idea that this new business wants to do they're going to make a new light bulb or they're going to make a this is funding for a small organic farm or whatever the situation might be i might like what they're doing and i might see value in it for the community again this is the investments are not related to profit so i'm not i'm not thinking the way i would think in today's world in the native economy if i have money in the native economy i'm thinking where can i put my money that will return to me the greatest gain personal you know my personal gain and finances in this system i'm asking since since personal profit is not part of the picture i'm asking where can i put my money that will receive the greatest gain in terms of fitness scores and community well-being and quality of life for myself and my children so those are the kinds of topics that become pertinent then when when the personal profit motive is taken out so um so again in relation to the native economy the native the government the say the us government also makes subsidies to various businesses but here all these choices are deeply democratic you know given that incomes are more or less equal everyone has more or less the same amount of money in the cbfs same amount of currency so everyone has a is essentially equal say in which which businesses get funded which businesses what you know whether they get subsidies or whether they get donations or whether they get loans and the reason for subsidies rather than loans and donations might be that um well that for one thing you have to pay back a loan so if you if a community wanted to support say a bakery and they thought that um you know this bakery is really great they don't they're they're not making a lot of money so they're not going to be able to really pay back loans very well but they're maybe even they're operating at a loss but there's just such a great community resource and we just love this little bakery so much that we would like to give this bakery a subsidy so that it can continue to operate even though it's losing money every year you know that would be one example of a subsidy or there might be we want to subsidize a hospital or we want to subsidize any any

operation that perhaps is not actually making money and couldn't pay back a loan on the other hand maybe some business is doing well and we would be willing to support it but we want to give it a loan instead of a subsidy donations really apply more to nonprofits and subsidies really apply more to businesses coach give a thought on that john all right yeah yeah it's like if uh a person who's granting money has ten thousand dollars and they're deciding whether they want to be the one who gets to decide where the 10 000 go or whether they're going to give it to the cafe who's going to spend it potentially at other local businesses and allocate it a different way and the employees are already taken care of and it's going to contribute to public good especially a locally um defined one and so it's like yeah i'll let them spend my ten thousand dollars that i was just going to give away anyway right right right yeah so so that raises a really a good point that i wanted to make is that what's really happening here even though it's currency that's circulating what's really happening here is just it's just a way to ensure that everyone gets the same amount of votes in this economic arena and what people are voting for is not so much money if money isn't the resource the this as especially in a mature leader the money is circulating as all as i'll show later so it's so it's not really a shortage of money like how will we get the money for this or where will the money come from it's not that it's using money as a voting tool to allocate other resources like if i give a um if i support a bakery say with with a loan or a subsidy what i'm saying is i'm i'm voting for human resources of the community to be spent you know in this bakery i want this bakery to employ these people and i want the bakery to provide this service to the community right so i'm voting for the service and the allocation of resources that are embodied in this bakery or it could be a hospital or it could be a could be anything but that's what we're really voting for is how we spend our time what do we what do the humans and this what are the members in their in their life and their 24 hours a day and seven days a week or whatever in their year how do they spend their time how do we want to spend our own time what jobs do we want to work in as a how do we want to spend our time do we want to be employed by non-profits do we want to be employed by regular businesses do we want to be employed by principal businesses this is a the evolution of this economy is the is the decision-making process of a community to decide how it wants to spend its time and where it wants to put its focus and how how what kind of conditions it wants to operate in okay all right um so that's a little bit of an introduction to the flow of tokens and dollars together um uh maybe i will say one more thing before we get off this is that suppose that um and this might be obvious from things i've already said but i just want to emphasize it but suppose that me suppose that me and a group of people in the

cbfs decide like hey it would be really cool to have a bicycle manufacturer in our town so so together you know the the 12 of us or 50 of us or the 100 of us decide like yeah let's support this bicycle business now every it's natural that not all businesses are going to make it you know some are going to fail and that's natural and good a community would want to take some choice some chances and fund some things that maybe even you know the reasonable chance it will fail but if it doesn't fail it would be great so so we fund a bicycle business and it fails after a few years okay fine the same money is still circulating in the community nothing's really changed now with what has changed however is the the allocation of human resources that were allocated to the bicycle business now can be allocated someplace else because the bicycle business doesn't exist and in the meantime those employees might become then not in workforce members right they maybe don't have a job for a short period so they go from the unemployed category in the persons or person's box to the unemployed and not in workforce box and fine but you know their income doesn't really change so they were you know they're not out in the street and neither is the business owner the guy who did the whole thing he's not out in the street he's his income hasn't really changed too much especially if he's a principal so he's fine the employees are fine everybody's taken care of the community took a chance they said hey this would be a good experiment let's have a bicycle shop it didn't work out but now we'll just re-allocate those resources to something else some other decision so you know the the funny that when i try to explain this to people the the and i'm using terms money and currency it's really it's difficult for some folks to get past their interpretation their the common interpretation of what money and investment and you know currency means and uh this is as i said earlier this is a lita economy but it's not like any kind of economy we're we we think of this is really just a decision this is a cognitive system in short this is the description of a cognitive system that a certain flow of is uses what i call currency if i can make a point on that john it's like if you use the metaphor of money a lot of times people debate uh characteristics like fungibility or store value or transfer value but if we think organismally and we think about let's say blood flow or neural information or hormonal signaling like the goal of the muscle cell is not to accumulate insulin it's to use it as a cue right to know when to change and so if you think about inflammation flow blood flow no cell needs to hang on to the red blood cells and capture things around it because it's part of a bigger plan so i think that that's a really nice attribute you captured yeah excellent analogy and that's very much what the leader framework is like it's kind of like a body kind of like an organism that that requires the different cells in that organism to receive blood and food and information and other things like that so and also

on that point when i say money is used as a voting tool even more broadly it's used as an information tool so the community is exchanging information when they exchange money uh uh still on slide eight uh i've explained now the blue lines the token and dollar flows but i haven't explained the green lines so uh this this the new club in this model uh as in as in all models for new clubs remains part of the larger world it's there's no the club is not segregated from the rest of the world uh apart apart from the rest of the world it is engaging in all the normal stuff it's paying taxes to the government it's purchasing products from member member organizations but also from non-member organizations and amazon.com or you know whoever so uh you know it's it's participating in the normal way and this lead of framework or the club framework is another layer of organization on top of the normal world um so these green lines are showing how dollars are flowing and again this is a simplification of the real world but it gives the idea of still so persons uh are paying the dollars in taxes paying their taxes and dollars i should say and the government is also giving some income support to unemployed persons also happens uh the government is giving dollars in the form of grants and subsidies and contracts to local organizations and local organizations uh i and i created this as a kind of a simplification but local organizations are exchanging goods and services with the rest of the world so dollars leave the community when an org local organization for example purchases supplies from some far distant and at the same time organizations outside of the club outside of the county are purchasing things from the member organizations too so uh yeah so that's that's a simplified version of how and obviously now we got a problem because uh you know we have our first challenge maybe you could say one challenge is to somehow design this system so to reach the aiming number three there that income is becomes more equal over time but now we have the problem we've created this this local currency the token and we have to pay taxes on income to the government and we can't pay taxes and tokens the government's not going to take it so we got an immediate challenge of like how do we make this thing work so that every individual has enough dollars to pay its the taxes it needs to pay and organizations have enough dollars so they can purchase you know goods outside the county in and yet and yet the whole thing can grow and and and reach its the desired aims so how does that happen another question designed the question is how do you make this thing good enough that anyone wants to join like fine i've drawn drawn this picture here of you know conceptually how it works but if i if i don't receive anything materially from joining why would i join um now there in actuality there might be lots of soft reasons to join i might feel like i'm more a part of the community or i might feel like my voice is heard more or

i might feel that uh you know by joining we're able make better decisions together about things but in this simulation i wanted something a little more substantial than that so so i in the simulation i made the rule that a person will join a family will join this if it increases their income and they won't join if their income doesn't go up so uh that's that that becomes the challenge then how do you inject enough tokens and get a good enough circulation of tokens and dollars in the in this membership such that my my income goes up from joining or at least most people's income everyone that joins their income goes up and uh how then how many people have to join to make this thing work like two percent of the population in 30 years or 50 percent of the population or 90 of the population like what what does it take so these are all the questions that we're going to dive into next uh uh first a few assumptions just to explain a little bit how this whole thing works uh the not in workforce population is static i think it's whatever it is 30 36 or something like that you know there's just a big as in the normal world a big chunk of people don't participate in the workforce they're either they're they're disabled or they're staying home they just stay home and don't work or whatever reason um uh i chose a seven percent uh let's see i think i think the unemployment rate starts at about seven percent something like that um uh everyone in the county is uh uh adults and grouped into two-person families just for simplicity so household income and family and computers anonymous the purchasing power of the token is taken to be equal to that of the dollar so when you hear token think value of about a dollar and uh structural unemployment is one percent so you know you can't get much below that no matter what you do and uh all employed persons work full-time you know just a few assumptions to make this simulation and again later some simulations could be more complicated all right give a thought there john yeah uh there's almost two extremes for the one is actually something that was tried in my little town of davis california with the davis dollar which was a one-to-one u.s dollar it was a stable coin before there was stable coins and so then they pegged the local currency it didn't end up continuing um it's not active to this day but it was a pegged coin and then the other alternative would be like a token that essentially people could they could make an nft out of it or they could go full crypto and they could inflate it and speculate on it do all sorts of leveraging if it was really being traded on the open market so there's kind of a continuum between a total protected in-group currency something that's pegged to a all these other options for the tokenomics so it's really um it's it's an interesting starting point but i hope others when they're viewing this see that there's a lot of space for experimentation yeah there's great points and um yes there's a tremendous amount of space for experimentation um and and on that note i want to emphasize that i'm thinking of

this as kind of a prototype concept is just to generate ideas just to give an example the goal of the of the r d program if it were funded is that it would it would evaluate and help to evaluate evaluate facilitate the evaluation of any number of designs and any number of design components that maybe are interchangeable or can be mixed and matched so this is just one concept and i would hope that over a period of years many concepts are um are tested you know obviously tested first in simulations and in in bench scale studies and things like that and then if the ideas are really good then also in field trials in real communities now having said that though i do want to emphasize that again this is a science-based evidence-driven program so so it's not just a question of hey what do you guys want to do well we want to do this and what do you guys want to do well we want to do this there you actually have to show that your design is good or better than others or how it can at least you at least describe how this design compares to another design in terms of the fitness metrics in terms of fulfilling the purpose of a system so there's a general general agreed upon a concept of system fitness and system and fitness metrics uh how to there's a generally agreed upon uh path to assess a system and then any design that is you know that is uh that is good relative to to the fitness metrics and and fulfilling its purpose then you know this is that's the idea like any conceivable design that improves fitness is a good one right that's one point now you also just on the currency i wasn't actually going to talk about this but like you brought it up so here we go you i think you mentioned two kinds of currencies one is kind of a potentially speculative currency like bitcoin another is a currency backed by the say the us dollar or something you know like you have everyone has to put in a dollar and then they get a dollar out of local currency my concept for the leader is neither of those so uh the the concept here is that the leader is um there's a word for it it's escaping me right now um it is just simply created out of thin air this is it's an accounting system so in the same crypto thank you also crypto is by fiat too it's by computational fiat yeah yeah so this is by no computation though this is just fiat but because we need more current we need more tokens so this is an accounting system and obviously there's a limit to how much tokens you create because the the concept here is that the token holds the value of about a dollar so if you created 10 billion tokens and flooded a community on day one the the token would be worthless right it can only be created to the degree that it can be engaged with and employed in the circulation it's all about circulation and in particular it's not speculative so you cannot sell in this system you cannot sell your tokens to anyone else that's because i in this system i wanted people to be uh you know engaged you know this is like a voting system i wanted people to be engaged in the voting or community choices and it

just wouldn't be right if i sold my votes to some other third party and gave them a whole bunch that just doesn't mean i can't uh uh entrust like in the cbfs for example there might be a there might be a community organization that says like an investment group in a sense a social investment group that says hey we've been watching you know we pay a lot of attention we we model the success of businesses and the needs of the community and if you entrust tokens to us and dollars to us we will choose good investments for them that we feel are the best and you know improve the community well you can do that that's fine you know that's all great but you can't sell your tokens to anybody and obviously then they're not backed by any resource dollar or otherwise but that's just that's just this design so as you said you know other designs would work differently and you could model out which ones improve the fitness of the community uh best both current fitness and anticipated fitness all right number slide 10. yep um so some additional notes um as i s as i said earlier one of the aims of this of the simulation was to show away a logical way in which incomes can grow and equalize over time but another way to say that is that elita essentially pays people to become members i mean that's how the simulation works if their income goes up by becoming a member then they become a member um so you know it's that's a that's a good selling point if if if elita has started up in my and i realize that by joining my income goes up well you know that's a good reason why i might want to join um and another thing is that we'll talk about three constructs here that's kind of the machinery that makes this whole thing go it's the income target the token share of income and the wage options so we'll be describing those shortly here um i already mentioned that a leader can create as many tokens as it can productively use limited by inflation concerns the simulation goes for 28 years the first 15 of that i term the growth and this is when the number of participants in the community is rising the number of tokens generated by the community is rising uh the the club itself the leader is expanding in in all ways and after it reaches its its peak at about 15 years and as it turns out that's the 90 percent of the local population end up joining the leader again because their income goes up um and then the following years the 13 years following that are called called the post-growth and that's just equalization period where everything kind of settles out in time and reaches a kind of a homeostasis point uh let's see here oh uh and on the right with regards to the leader we can talk about family income pre and post cbfs right so so if you're if you need if you're required to contribute a certain fraction of your income to the cbs well then obviously your pre cbfs income will be higher than your post cbf cb fs income uh when i do when i do comparisons here i'm really making comparisons on the post cbfs income so you can think of that as really

pre-tax uh take-home income all right um and maybe one last thing to talk about oh go ahead one funny note on that is depending on the uh governing tax body and how they regulate whether a certain kind of crypto transaction is taxable or not every transaction of a token could be considered taxable i'm not a lawyer obviously and this is the future but i'm just saying that depending on what kind of overarching tax structure exists it could completely make or break this total structure for making it tax-free to making it something that grinds to a halt oh if if uh if this was tax free the token was tax-free if i could get income as tokens tax-free then this whole system is going to work like a charm i would say i've designed it under the assumption that everyone will have to pay taxes on the tokens that they receive so uh that's the more conservative assumption and it and it holds true and we'll show how this how it all works out um i do want to uh emphasize that this that the you know the the the kind of the structure of the and the way the cbfs works and the contributions to it and things like that it's not dependent on the good will of members these are the it's these all occur because of the rules of the club so if you want to become a member here are the rules the rule is you know the you know there's a variety of rules that you follow and once some of those rules are encode what your contributions to the cbfs will be again once you put the money in there then you get to choose you know largely you get to choose how that money is spent all right on to uh eli yep so i mentioned that there's uh three constructs that sort of drive this whole thing and the first is the income target i also mentioned that simulations would be available long before the first field trial ever ever occurs so in these simulations the community that's being simulated would choose an income target and that income target is a is a increasing level of family income up until up to a certain plateau and it levels up so that path the expected path of that increase in income family income is part of the design of the leader and it's and different communities could choose slightly different uh uh um functions or paths to get there but that's the concept that that everyone knows in advance how we're gonna get to this uh plateau level you know how many years it'll take at what rate for example like that um a second uh construct is the wage option and uh this is to address a certain problem so for example i belong i'm in a family my husband and wife team and i make a lot of money and my wife is not working say as an example well then she could join the leader and be included in the not in the workforce group and she could get a high income just from being a member of the but i'm going to stay outside the leader because i'm making a ton of money anyway and i don't need the leader you know so best choice for us would be i i stay out of the leader she joins the leader and our income goes up considerably because of it but that would be a little bit of a freeloading uh situation and

would not be good you know it wouldn't be good for the lead as a whole so to prevent that kind of thing there's two wage options if our family income is of a is as this uh income target rises in any given year uh if our income target is above it or if our income say i'm working in some business if our income is above the current year target then um uh then we choose wage option and the and the club will give us a stipend you know above our normal income in tokens a small stipend so we get a little bit of we're doing well we have a lot of money because our income is high and we join and we get a small amount of tokens that we can play with in the cbfs and stuff on the other hand if we our family has a low income lower than the target and we join then we receive the whatever current year income target is so in the first year that might be low but each year it increases uh monotonically and by the time the simulation is over everyone essentially is choosing this wage option number two and is receiving the um receiving the current year income target at its plateau that's a little bit about that and then i i should say that the uh i the the set of families whose initial income is is below the 90th percentile that i consider those the target population and as we'll see the leader uh increases incomes for 90 of the population and it's that 90 that joins the leader by the end of the and there's also a token share of income target tsi target that is in any given year um you know how are the dollars and tokens uh you know what is like if i'm going to purchase goods and by local community and local how many tokens am i going to give relative to dollars that's also a design parameter okay slide 12. one question is you mentioned a two-person household where one person had a high income and one person didn't um and then you said it'd be free writing if one person joined and the other didn't but what about one person wanting to join who doesn't have an income i mean where does that play in oh oh oh that didn't play into this simulation because for simplicity i assume that everyone was in a two-person family obviously in a more sophisticated simulation you would have young adults that are single and you know and so on so that the simulation just doesn't have that level of sophistication but i would i would argue though that i'm showing by the simulation the main concepts and that the main concepts make mathematical sense so i i don't think it would be all that difficult to have a bit more sophisticated simulation where there's actually one-person families and things like that but good point um okay so i'm gonna in a moment i'm gonna show you some of the results of the simulation and just keep in mind that that's a simulation for a hundred thousand people and i'm using uh for initial income values and income distribution i'm using data u.s census data for 2011 from for lane county oregon from the u.s census so that's that's how i'm making this semi-realistic in part is i'm trying to mimic the distribution that was in lane county

oregon in 2011 and then and then that's to start with year zero and then every year it changes as it changes and if you never become a member then your income stays at the same level not obviously not in not considering inflation in this model it's just a simple model and so we're kind of explaining the concept of inflation for the moment and it could be easily added in a more sophisticated simulation um uh as far as other ways to make this semi-realistic i as at the bottom of the left-hand side there i choose a tax rate that is a reasonable tax rate there's a bunch of other uh parameters that i chose to be reasonable parameters of what i could you know what's a reasonable tax rate what's a reasonable tax rate oh i chose uh for individuals i chose ninety percent of adjusted gross income you know and everyone gets a 2500 deduction and you know things like that so like in the ballpark what that's what i'm arguing is that these things are in the ballpark um and again a more sophisticated simulation could look at a broken you know like a more complicated tax rate or something um i also look for example like you know what what appears to be the average rate that people donate to nonprofits just in dollars just regularly and you know i chose a percentage of income for that too things like that just to just to kind of get semi-realistic and it turns out that the simulation is not very very sensitive to those parameters so they could be changed a bit and the same the simulation should show the same results now on the other hand there's some token exchange system specific parameters and that they definitely drive how this whole thing works there's the uh some of them i mentioned for the constructs there's the the formula is how how uh the income target rises every year uh that makes a big difference and and so to some of the others the the amount that people contribute to the cbfs makes a difference and so on uh and speaking of a month they contributed to the cbfs there's different kind of categories of the cbfs there's the nurture category for example the subsidy category the loan category and so on and there's what i call ear marks for all of those in this simulation like a certain percent of income goes to group and cpfs certain percent goes to lending a certain percent goes to nurturing et cetera i called it nurturing by the way is because that's i wanted to convey the idea of the community is nurturing itself through this process it's it's providing votes to those people that wouldn't otherwise receive votes from like from their employer so again money is a voting tool uh okay all right initial employment rate is seven percent unemployment rate is seven percent uh and the initial fraction of employees who work in non-profit organizations is now uh when uh when uh when a community say the leader existed and the community was getting together and kind of form forming to use elita they would have to make all kinds of decision like what kind of income target function are we going to use how many years do we it'll take to reach

our you know full capacity uh there would be a whole bunch of decisions like that to make and uh one of them would be um how many what kind of makeup of the our like makeup of the of organizations do we want do should we all work for nonprofits like what if we what if a hundred what if 90 of us the membership actually worked for a non-profit that's kind of an interesting question you know i and i would argue that this will work the community could choose like fairly extreme values and should work as long as the system continues to function as a cognitive system and as long as the community is using it to solve their actual problem so if the community wanted to to have most everyone work in the nonprofit sector and wanted the nonprofit sector grow food for the community and provide services to the community it could do that i think it could do that if it if it instead wanted to fund principal businesses to a large degree and nonprofits to a smaller degree it could do that so this again this is a this decision making process is a way for us for a community to decide what it wants to be what flavor it wants its economy to be where where it wants its individuals to work like you know being a member of a community like this i would want to choose jobs that i that are useful to me and to others and that i would want to work in you know like whatever whatever those would be but an interesting thing though is if if a chose to fund a large part of its workforce in non-profit sectors it could have say you know a large number of biologists out of repairing forests and street and streams and you know it could have a lot of forest workers uh doing you know repairing things there could be a substantial amount of ecological reject narration funded from this but there could be other things it's whatever the community feels is going to increase its fitness you know it's its capacity its well-being its quality of life the community can choose not only which jobs to fund but what kinds of jobs what kinds of organizations all right all right uh on to the next slide 13. so this is the way the simulation occurs and it goes uh in annual steps and in each in each zone each year the first step as new members are added according to uh expected participation rate and uh and some of those people who are already working in what i call lead of funded new jobs l f n j uh that is you know the lead of funds an organization but that organization might go bust and then those people would lose their lead of funded new jobs and they switch over then to the to the unemployed or not in a workforce category so uh number two every year some people lose their jobs some members lose their jobs uh the number three the cbfs makes nurture payments to those people who are not in the workforce who are unemployed um uh i'm just gonna i'm not gonna go through each of these but i will not mention number five um so the leader has injected tokens into the economy you can't pay taxes with tokens so so by definition some dollars have to come from elsewhere

another factor is that in a poor community it may have a shortage of dollars relative to the rest of the world right other other other areas in the country might have a higher percentage of dollars in circulation so somehow this club this leader has to pick up dollars from elsewhere in order to fund its uh you know in order to achieve the level of income that it's aiming to achieve and it does that by one main one main process and that's by having a kind of a sophisticated buy local program that is uh it's the community is supporting local businesses and um spending a good chunk of its of its resources in the local businesses and then by definition that money is not flowing out of the community it's staying in the community longer it's circulating in the community longer and uh it the community obtains dollars from elsewhere by uh by having it by adjusting its trade balance with the rest of the world so one of the one of the the the statistics that the community would be tracking is what is our trade balance with the rest of the world how are we doing are we are we adjusting our trade balance as necessary to bring in the dollars that we want and uh in this simulation as you'll see the the the families they have the families actually reach what would be considered very high income from this the 90th 90th percentile of starting income so they families would receive quite a very high income and that would include tokens and dollars and as it turns out the dollar portion of that is about the national average if you looked at all all incomes in the u.s that community ends up in a sense obtaining its fair share of dollars so if every community in the country were to do this there wouldn't be a shortage of dollars because those dollars would just be more equally dispersed everywhere and that's kind of important you wouldn't what you would not want to do inside d program is to design a system so that small community can do exceptionally well at the expense of everybody else so that's that's not what this is um so uh number five was that you adjust the trade balance as part of these steps every year and again this is a very simple simulation so at the end of each year all the individuals spend whatever has gone into the cbfs or to taxes or for other purposes like that they spend that in local and finally number seven individuals may pay taxes to governments in dollars yeah so that's it so that's kind of the steps you get there that's how it sort of works mechanically here's some of the uh parameters that i used over on the left are the kind of general parameters uh excuse me those are the the targets for the for the community so the income targets both starting and ending values uh and this is for the growth period which is a 15-year period so the uh income target starts low at the essentially at a what would might be called a minimum wage right and then it rises up to uh over almost 110 000 tokens and dollars 110 000 family income in currency and that is the 90th percentile of starting values so by the way i've already said that

every year the the club essentially pays people to join so uh obviously in year one it's not asking anyone to reduce their income down to the minimum wage right so no one is asked to to lose income in uh people join because their income increases now some people might join because they just want to be a part of the community part of the leader and their income would join their income would increase too if they're above the income target that year their their income would increase by a small stipend paid in token so even if i'm making a million dollars i can join and i can receive some tokens and i can participate in cbfs and other things like that um so that was the table on the left and the table on the right are some of the ear marks that i use in this particular simulation and you might notice that the highest ear mark there is for the nurture arm so a good a good chunk of my kind of raw revenue come in my raw a good chunk of that almost 40 percent is going to the nurture arm of the cbfs and that's pain again for anyone who's uh unemployed or not in the workforce and uh other than that so some of my income goes to the lending arm some goes to principal businesses some goes to nonprofits goes to subsidies of various kinds goes to donations for nonprofits and uh and so on and i also have obviously if i'm putting money into this lending function then i'm building up a kind of a savings account in the cbfs that if people are paying back my loans then every year i'm maintaining a a good chunk of money in the cpfs itself kind of like a savings account i think uh yeah that's about it there's a couple of rules of thumb that i realized in trying to find uh parameters that actually make this thing go and the one rule of thumb is the the nurture arm has to have a high enough air mark so that that actually functions as as it's supposed and uh and also the donations earmark for local nonprofits has to be high enough so that any non-profit that i fund can actually pay its employees the the the the income target right at least the income target by the way the principal business one of the one of the things that demarks principal business is that it pays no higher than the final income target so the final language income target is about 107 000 token tokens and dollars and uh a principal business would never pay someone more than that and ideally most people who join a principal business would become members and i think they all would become members and participate and delete it itself so uh so that's that's a few things that distinguish a principle of business and obviously a community would want to fund principal businesses because this is a business model that is designed to support the lead and club model itself so it's it's designed to support the community and if i were voting in the cbfs i would really you know like i value being a member i value my community and i want my i want this thing to work and i want i want to receive the benefits that this system is promising and the and the the way to get there or one

way to get there is to mostly fund principal businesses so that they are serving the community in the way kind of like a non-profit service and they're paying wages that are consistent with how this whole thing works and they're really being responsible to via this business model and it's really in my best interest to support principal businesses with you know subsidies and donations and loans and things like that not that not that there's no room for for normal for you know regular businesses there is you can fund those too but you know by and large you want to put a little extra emphasis on principal all right um yeah that's it for parameters next slide now we're going to start looking at what actually happens uh we have about 20 and we have about 35 minutes left i think we're going to get right through the end of this so i may have to go a little faster as we as we get to the end so the first slide on the upper left is uh the fraction of the county population and you can see in that blue line that's the that's the the rate at which people the participation rate which is decided in advance you know this is what we expect to happen and uh there it goes straight up and then it reaches the plateau after 15 years so that's after 15 years 90 percent of the county population has joined the leader and they've done so because their income has gone up uh the the if i were if i were if i were a member of a citizen in the county and i didn't have a job well for sure i would want to join the leader because i'm gonna get my income is gonna go way up if i join the leader relative to not joining the leader so uh i account here for every not every unemployed person or not and workforce person to to to actually join the leaders so they can be part of the you know receive funding from the nurture arm of the cbfs um down below that there's the income target the income target the blue line again after 15 years it reaches a plateau of about 110 000 tokens and dollars and then it levels up and and again i want to emphasize that at that level every community in the country could do this this is not preclusive if one community does it that doesn't mean that others can't every community could achieve these levels you know based on the assumptions of the model on the right is the fraction of income uh over the years so a fraction in the let's just look at the the green line the green dotted line is the token share of income target so uh in the early years that's a very low fraction most of the income is in dollars and then it raises over it rises over the years to eventually reach about so at the end of the simulation about 35 percent of incomes and tokens and the rest is in dollars um and then the fund starts with the bottom right graph this is the average family uh starts at about forty thousand that is you know from this from the census data and starts at about forty thousand and you can see what with the blue line over the period of 15 years it uh more than doubles so family income more than doubles it reaches the 90th percentile of starting the red dotted line is

showing the similar thing only it's including the money that people have in their savings account in the in the cbfs so there's a two-person family each is contributed to a maximum of 30 000 to the savings account so on top of this high income families also have sixty thousand dollars in savings in the in the cbfs um you might notice that uh you know the the all the growth is happening during the first 15 years but you can tell from the blue line the family income line at the bottom graph that it there's a little bit of a lag it takes a little bit of time for the whole thing to actually reach its maximum family income and that happens about your 28 or something like that i might also mention too that the i've just tried in this simulation i was just trying to find kind of reasonable parameters and and a growth period and such that it would all make sense and would work and we're not asking we're not expecting the community to change on you know in year suddenly everything changes it wouldn't be like that it would tend to grow slower but having said that you could think of the simulation as what might happen in the first leta but imagine that there's already a hundred liters or 200 or 500 liters around it's really clear how these things work there's lots of data coming in from leaders all over it's clear that incomes are rising and interest by the general population is you know growing very rapidly like suddenly it's the new thing and everyone wants to join because everyone's income goes up and empowerment goes up and their sense of community goes up and they're and they're aligning their purpose with the you know some greater good that's it feels good you know there's a whole variety of benefits that come from being a member so so now a new so someone wants to start a club in in another city and suddenly there's a tremendous interest on day one well then all of this would go a lot faster it wouldn't necessarily take 15 years it might go you know in half that time or whatever uh on slide 16. on the left is the unemployment rate as i mentioned it starts at about seven percent in the county and you can see that uh after about 15 years uh the county achieves full employment so uh that all works well the lead is generating jobs and whether they're non-profit jobs in the nonprofit sector or for-profit sector the unemployment rate everyone's being engaged that wants to be engaged now keep in mind that a certain percent stay out of the workforce and it just shows a typical percent whatever that was 36 or something so that not everyone is working some people are doing other things in non-formal jobs or whatever term you might want to use that you know everyone's engaged somehow people are actively engaged in their in their lives and in their community but some people who are in the workforce the employment rate is very low and and the unemployment rate in the club itself uh reaches its lowest level after 10 years it just takes a little while for the rest of the county to catch up on the left this is uh this is the

before and after snapshot of what family incomes are uh the upper graph shows the distribution from the census data on day one or day zero and i uh the bottom graph shows the same thing on the 28th year of the simulation and you can see that essentially everyone in the county uh 90 of the people in the county are making the income target which is just over a hundred thousand so so it goes from a highly unequal uh income to a almost completely equal income over the period of 28 years let me let me ask a question john so i i still see that between 350 and 400 there's still the uh 10 who don't join uh ostensibly because they would stand to yeah right so how do you take that's great they don't they wouldn't join a big oh sorry just how do you take the other and make their average value something with fewer people working productive jobs having a higher average than 80 percent that's just below 100k in the top graph um if i understood the question it's kind of what do you do with the night with the 10 who don't join and do is there some kind of do their income somehow equalize over time or do they like what well let me just ask again because they they don't change so ignoring the ten percent whose income does not change how do you take the um belt the bell curve you know right side here which has an average below 100 000 and then by reducing the variance also increase the mean while potentially having fewer people with income generating jobs just how does the extra usd per year come into this system okay again i'm not sure if i understand the question but how does the extra usda us dollars come into this system that's through balancing of the trade deficit so uh that's how that new dollars come in that weren't there before right so what the what this club is doing essentially is adjusting its trade balance with the such that it's getting its fair share of and uh you know the the dollar income of the club equals the average dollar income of the of the us at the end of this simulation did that answer your question uh so it's predicated on basically a locally positive economy that's generating a trade excess to get more usd because you need more usd to pay that number of people a hundred thousand versus between zero and a hundred thousand and so even though fewer are making potentially productive jobs fewer are not in the workforce there's so many more usd coming in that we can raise the income for them and everyone else i'm just saying that that that's built in right that's built in yeah that's all built in yep yep now what i thought you were originally asking is what do you do with the other what happens to them we'll talk about that in a moment but i'm going to argue that their incomes actually fall towards this um this plateau of around 110 too i think social pressure social pressure would ensure that that would be one that would be one thing and there's some other reasons too um well we might as well talk about it now since we're on the topic so uh so who's so so let's suppose that i own this uh some corporate i own a

franchise of a corporation i would say mcdonald's or something could be anything and uh i'm you know i take in money from the local community they spend money to buy my food but then i'm sending money outside of the community because i'm paying you know franchise fees and other things corporate fees and other things so some of my money's leaving the community i'm paying low-wage workers and that that's what makes this whole model work is that i can get low-wage workers to you know sit behind the counter and make hamburgers or whatever now what happens when there's no low wage workers to choose from like everyone's joined the leader everyone who i would have paid thirty thousand a year forty thousand a year is now making a hundred and ten thousand a year where does my workforce come from how is my business model going to even function if i don't have a low low income pool to draw from so i i'm going to my business model is going to be strained to say the least uh it won't work very well it's changing the game right it's changing the foundations of how the normal business works in the world um so and not only not only would people not be available because i because the leader would be paying higher incomes higher wages but but those people should be enjoying their jobs a whole lot more they would be they would understand that they're helping the community by working in the bakery or wherever they're working it's part of a community effort to raise the fitness of the community right like we're all in this together and it's and it's and i'm getting a voice of decision making by becoming a member so there's lots of reasons i would want to be a member and very few reasons i would want to be a low-paid worker at you know at mcdonald's or anywhere else and that's that's another reason and the third reason is there's the ip commons that i mentioned earlier um so there's all these you know the leader has now funded all these new businesses some are technical businesses some other businesses but there's all kinds of property being generated in particular intellectual property that that is shared amongst the membership amongst the club and maybe even between clubs so suddenly there's this enormous advantage to to principal business or to uh to member businesses to becoming members because you get this you know this volume of a free intellectual property and standard businesses that are outside of this do not have access to that to you know they have to pay pay royalties to get that you know ip stuff so that's another reason uh and i think there's actually still one more reason that i can that i had i want to talk about well a successful community a successful community would have many leverage points and many ways to draw income in and an unsuccessful community would have no one wanting to help them except for the us government paying usd which i also saw in the model right that the usd is providing by the government so it's like welfare at the country level and

also at the community level yeah yeah so imagine that's a good point and imagine how that's in fact say a county government county level governance or state level governance or even federal governance what if it didn't have to pay like all this unemployment insurance and all all these other things fees that the government has to pay you know when people are when people need help right all these programs and the and all of it is being taken care of at the community level by leaders and clubs like leaders so the so there would be actually like in that sense you would expect that the local state and gov and national governments might actually like this concept you know because it's reducing their liabilities you know that's especially at the local government the local government as i said the local tax revenues are double in this in this simulation so you know the county government might go like hey good idea this is if this could work this is a great idea and the individuals might say hey this works this is a great idea but my income is going to double uh great fantastic uh just a bit on an income obviously with this the lower your starting income the more you benefit from this so if you happen to be someone at day zero where your family income is fairly low yours is going to have the greatest rise compared to someone who's already making say mid-level income or something but it doesn't matter it is like the rising boat catches everybody by the time it gets up to them up to the plateau after 15 years uh some more results on slide 17. uh on the upper left hand corner uh you can see how much money is circulating in this uh cbfs and um i i want to emphasize again again that this is a community of a hundred thousand people this is a small community right so this small is now circulating six billion in annually that's six billion that it has to choose about like what jobs do we create what what what nonprofits do we support what do we want what kind of parks do we want what is the road quality how's our hospital system whatever decisions this community needs to make it's got six billion in currency to play with so uh an enormous amount and that should be enough for the community to make whatever choices it wants as far as like what are we going to do how are we going to get there are we going to improve our quality of life and maybe just on that note i will mention again we've already said this there's no club that's a that's an entity even in and of itself as clubs start to form uh and develop there would also be along with them networks of clubs that are forming and developing that are also a cognitive system so the cognitive system is playing that cognitive architecture is playing out at the local level at a club and it's playing out between clubs as clubs need to make decisions about how they're going to solve larger problems right so so again no club is in this for its own good and only its own good is this is a you know clubs are connected this is a greater good project um the bottom left these are how the

workforce gets divided over time so on day zero most of the workforce is in the regular state of businesses almost 90 97 is in a regular standard business a little bit is in the nonprofit sector and obviously zero is in the principal business sector but after after the period of 28 years it achieved some different balance of the torque force and i just want to say again that whatever balance that is it's a balance that the community chooses in advance right so this just didn't play out by itself these are based these are all based on community decisions all along the way like we want this community wants to achieve a fifty percent uh you know that to split its workforce uh fifty percent into the nonprofit sector or forty percent or eighty percent or ninety percent or whatever it wants um upper right draft are the donations made to nonprofits these are dollar donations apart from the cvfs and i'm just using a like a fraction of income goes to donations like a general fraction and you can see that uh donations to local nonprofits more than double so you know this at least in theory this whole system is like a boom for county governments it's a boon for the non-profit sector it's a boon for the education sector uh and then everyone else's income is going up and unemployment's going down too so like who loses the only people that are losing from this whole thing are the people who hold enormous economic power on day zero you know like those people aren't really gaining from this and if lots and lots of clubs were to you know lots of lots of cities were to implement clubs and this actually happened in scale you know distributed at scale then the real losers in a sense would be those who are quite you know quite wealthy to begin with like billionaires say or multi-millionaires because their income their incomes would probably drop as i've mentioned over time only because of social pressures and because they can't get people to work in their businesses and but at the same time everyone's well-being increases so it's not like anyone loses everyone wins here like the commute the world is you know all communities are more empowered to solve their real problems that matter this is a cognitive aimed at achieving sustainability and thriving into the future so there really isn't anybody that loses from this there's just some apparent losers you know and not very many of them at that um the lower graph on this i won't say too much i'll just explain the blue line there this is the number of tokens uh that are generated by the community so obviously in the early years you're for the first 15 years you're generating uh because you're growing in this growth period and then after that you have the tokens are just circulating and importantly they don't circulate outside of the membership so if i'm a standard i there's no one can oh that's the other reason why why that my clubs would and club businesses grow is that if you're not a part of the club people can't spend their tokens you know at your establishment so i i might be a local business and i don't

want to participate and that's fine it all goes as it goes but everyone else's incomes are increasing part of that income isn't tokens and those people can't spend their tokens at my establishment the only way they can is if i join the if i join the club then they can spend their tokens at my establishment uh page 18 and i think this is the last of the results i was going to show this is the results just for the local government this is tax revenues for the local government at the red is the tax receipts so for for this uh lane county is estimated that it starts at about a billion dollars annually in revenues and it more than doubles by the end of the simulation so their tax revenues go way up and you can imagine what counties would want to do with you know if their revenues tax revenues doubled you can imagine they could put that to very good use okay then just some final remarks on these last two slides um and this this first one is something that we just were touching on um what happens to the other 90 that don't join the leader right so i'm i would argue that in the long run even that 10 that has very high incomes would see their incomes move closer to the to the income of the target income for the leaders and that's for all the reasons that we just mentioned that that uh business owners don't receive the benefit of tokens because they can't accept tokens uh there's social pressures to belong to the club because it it's in everyone's benefit for people to belong to the club uh the ip is there's an eye by people that people have access to if they join and so like a number of reasons maybe there was one other that we talked about uh so you would you would think that you know this is the new you know like if this is happening that it's the it's it's like who who uses a typewriter anymore right hardly anybody but it's because computers work better for type setting you know this is technology that's another way to say that this is a technology this is a decision making technology and it's both a soft and a hard education is part of it there's you know there's all these components to it but it's technology and uh you know that's what happens in the world a new technology springs up it's it's far better than the old one and then suddenly the old one is toast well the old one uh becomes subsumed into conflict at a higher level so you could have different uh chains that are competing so uh the idea that still one part for me is that just thinking about evolutionary transitions in individuality it raises the level of conflict to another level up from what we've seen before with everything that politically and game theoretically would come with that and the second piece is um um i i don't update my priors based upon models without parameter sweeps so it's hard to look at a given layout and imagine but it's something that we actually use as a north star for people who are on that political journey they might see that as beneficial but as you're pointing out people who don't see it as beneficial are not going to join until the incentive structures are restructured for them

or that will happen yeah yeah absolutely yeah and everyone's learning from this so you know the communities the world is learning from this right and the world is learning over time whether this is even a good idea or not does this whole system work and obviously if it doesn't work if it doesn't produce these benefits that are that are that are expected in advance if it doesn't do that then it doesn't work and then no one would join because it's a stupid right or it has flaws that need to be fixed or something so uh yeah everyone's learning together and uh there could be numerous trials going on of different systems at the same time and you know try for the world everyone can be trying to figure out what works best in this what kind of systems work best and um yeah all of that's happening it's evolving as as we go it's evolving but the important thing is is that this is a framework to facilitate societal cognition you know at the local level and um if we think of our societal systems as cognitive systems then it's a perfectly reasonable question to ask what kinds of system designs work best and that's what this whole r d program is aimed at just asking the question of all conceivable designs which kinds of designs might most facilitate societal cognition and therefore achieve the intrinsic purpose of a of a society which is to thrive now and in the future so what and you can imagine all kinds of trials to get to help answer that question um next slide and this is the last one just a few remarks i've touched on this already but but an organism is agile right like i have among among other things i have legs i can run from danger i can do things i can change my behavior uh as the community we can we can change our behaviors we can we can do whatever we need to do to achieve whatever task is in front of us or you know at least achieve most tasks but in particular elita is agile like it is flexible it can choose what jobs it funds and suppose that you know 10 years into a into a new club some big thing happens or some new challenge that wasn't anticipated it just arose well this is decision making under uncertainty that's the whole concept of like this whole system facilitates decision making under uncertainty something new has happened now it has to the community has to respond to this new set of conditions it can change its makeup of what businesses it funds what organizations it funds who does what it can act it you know it can change the how many people work in the nonprofit sector versus for-profit sector it can essentially use it can vote to to focus its attention and its resources human and otherwise on whatever problem it needs to focus its attention on so it's agile by design and part of that by the way is is having a fairly equal income so in our native if i'm a billionaire it is really hard for the society to allocate my billions of dollars for some worthwhile project right because because of the you know the private property laws that's my private property and you can't use it right but if everyone's

income is equal then everyone is participating in that redistribution of attention and resources to whatever problem is important to the community so uh a community empowers businesses with with funding for example and it's important that it can retract that uh support when necessary so the the design of this whole system is such that there that things can change uh rapidly if need be and the community can redirect its attention to whatever problem it needs to address um again the next one there is that the main resource is not money it's human it's human attention and and energy and also to some degree natural resources but the real purpose of money is simply to to allocate human attention and human energy and resources into those problems that actually matter and and lastly again uh the leader can because of its agility because members are choosing what kinds of jobs to make what sorts of businesses what sorts of right what sorts of economy it wants it can choose any flavor of an economy it wants it creates the jobs that it wants to work in and the kinds of jobs that the people want their children to work in so it can do whatever it wants as long as it functions as a cognitive system and as long as it actually meets the essential needs of the organism of the community and that is uh it in a nutshell or not in a nutshell that's it in those long distances there's the the blueprint haiku for the nutcracker it is a few levels upstream but that's why it's fun to have these open-ended discussions and just allow um what you've worked on for so long to flow out just hear how you're laying it out and maybe in closing out this little discussion as we head into the next two weeks with the group discussions what how would somebody get involved or where can somebody come into play after seeing this or reading the papers oh well um i can be reached through the website principal.societiesproject.org and there's there's other resources there so you know one start would be just to familiarize yourself with the material a bit but we have a mailing list so people can just join the mailing list and as things progress and as opportunities arise then we can shoot you know we can use our mailing list to contact people um this our this whole series is written to the scientific community so that's that's my target audience for the series of papers and i'm hoping that i'm hoping that i can entice the scientific community to pick up this topic as a worthwhile topic of study and endeavor you know can we can we as a as a scientific community address these issues create designs test designs identify fitness metrics for community you know describe how community cognition works and what is how it works best and the conditions you know necessary are we are we ready to to pick up this topic and i'm hoping the answer is yes because it looks rich to me and on so many levels could be extremely valuable to society it and obviously it's aimed at societal and it's already quite clear that that different groups academic and otherwise generally understand that bold change

is necessary so this is one form of bold change now uh academics in particular please do reach out if you think if there's any number if you have some funding there's any number of small studies that could be done right you know immediately this simulation could be uh um expanded in a number of ways uh a study could be done just looking at just asking the public what they think of deep transformation and what they would think of of income equality or what that you know like laying out some possibilities to the public or if you're in media whether that's literature or music or movies or documentaries or whatever please contact me because just getting the word out to the public that transformation in this sense might actually be viable and possible that that's like half the battle is just to say we can do this there's a way we can do this this is open to us if we choose to take it and uh so that that's important there's so if you got any idea about how to what you'd like an idea of how to contribute to this effort then great you know send me note it's like either they can read it now and send you a note or hear about it later or not so that's where it goes john thanks so much for these awesome sessions though because it was great reading the papers and great to go through them again and we'll all be looking forward to speaking with you on the upcoming tuesdays at 7 a.m pacific so we'll talk to you soon great i really appreciate the opportunity absolutely john talk to